Topic C

Multiple choice questions

1 A pendulum undergoes SHM with period T. The mass of the bob is made four times larger. What is the new period of the pendulum?

A $4T$
B $2T$
C T
D $\frac{T}{2}$

2 A pendulum on Earth undergoes SHM with period T. What is the period of this pendulum on a planet where the acceleration of free fall is half that on Earth?

A $\frac{T}{2}$
B $\frac{T}{\sqrt{2}}$
C $T\sqrt{2}$
D $2T$

3 A mass on a vertical spring oscillates with period T when the amplitude of oscillations is x_0. What is the period when the amplitude is changed to $\frac{x_0}{4}$?

A $\frac{T}{16}$
B $\frac{T}{4}$
C $\frac{T}{2}$
D T

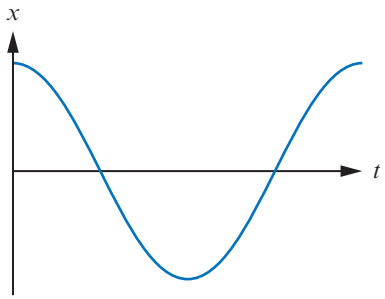
4 The net force F on a system is related to the displacement x from equilibrium. In which case will there be simple harmonic oscillations?

A $F = 2x^2$
B $F = -2x^2$
C $F = 5x$
D $F = -5x$

5 The frequency of oscillations of a mass m attached to a spring is f. What is the frequency when the mass is changed to $4m$?

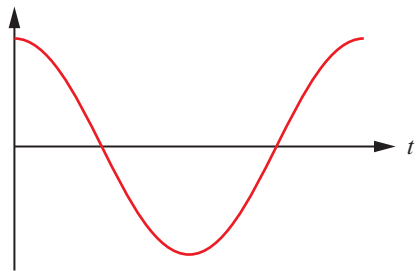
A $\frac{f}{4}$
B $\frac{f}{2}$
C $2f$
D $4f$

6 The graph shows the variation with time of the position of a body in SHM.



Which graph shows the variation of the velocity of the body?

A



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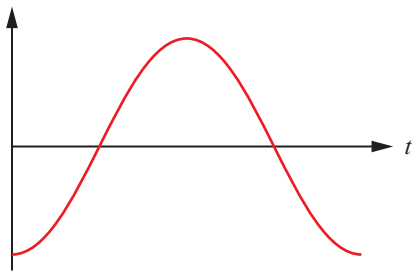

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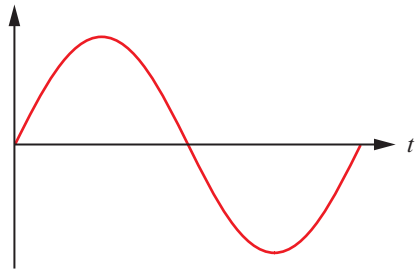

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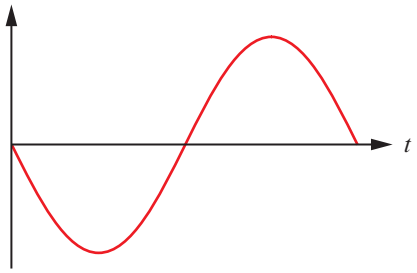


7 The acceleration of a particle in SHM when the displacement is –2.0 cm is +3.0 m s^{-2}.
What is the acceleration when the displacement is 3.0 cm?

A $\frac{3}{2}$ m s^{-2}

B $-\frac{3}{2}$ m s^{-2}

C $\frac{9}{2}$ m s^{-2}

D $-\frac{9}{2}$ m s^{-2}

8 Which statement is correct for simple harmonic oscillations?

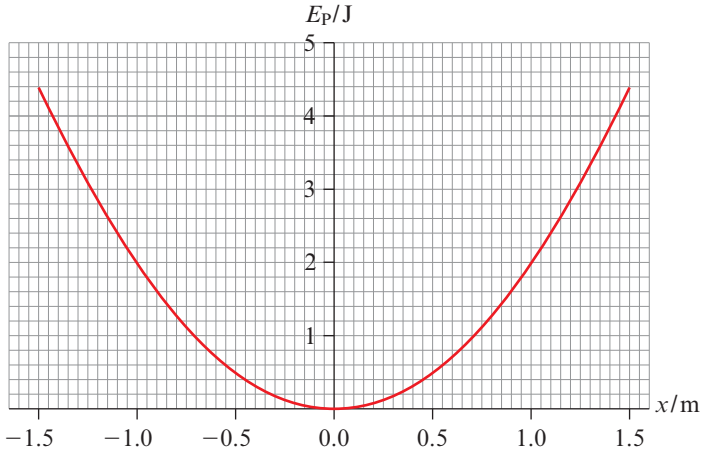
A The acceleration is always opposite to velocity.

B The acceleration is always parallel to velocity.

C The acceleration is opposite to velocity for the first half of the period and parallel in the second half of the period.

D The acceleration is opposite to velocity for a total time equal to half the period and parallel for total time equal to half the period.

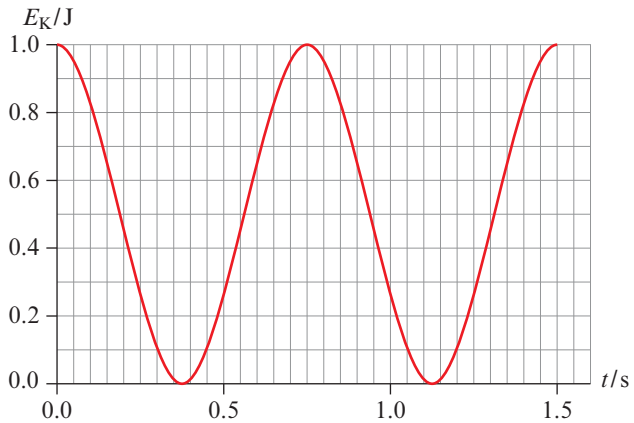
9 The graph shows the variation with displacement x of the potential energy of an object in SHM.



What is the kinetic energy when the displacement is 1.0 m?

A 0.5 J
B 2.0 J
C 2.5 J
D 4.5 J

10 What is correct about the kinetic energy and the potential energy of a body executing SHM at the position of maximum displacement?

	Kinetic energy	Potential energy
A	maximum	maximum
B	maximum	minimum
C	minimum	maximum
D	minimum	minimum

11 The graph shows the variation with time of the kinetic energy of a particle in simple harmonic oscillations.